

# KWANKHAO SIVASOMBOON

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## PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building AI-powered systems across computer vision, GenAI/RAG, backend services, and data pipelines. Developed real-time computer vision applications, LLM-powered automation, and deployable APIs using Python, FastAPI, Docker, and GCP. Work spans AI model integration, real-time inference, data processing, and backend engineering.

## TECHNICAL SKILLS

**Programming & Data:** Python, SQL, MySQL, PostgreSQL, Pandas, NumPy, MATLAB

**AI/ML & Computer Vision:** PyTorch, scikit-learn, YOLOv11, OpenCV, OpenVINO, InsightFace/ArcFace, ResNet-CRNN, CTC Loss

**GenAI & Retrieval:** Gemini, Groq, Ollama, LangChain, RAG, ChromaDB, Embeddings, BM25, RRF

**Backend, Cloud & DevOps:** FastAPI, Pydantic, REST APIs, Docker, Google Cloud Platform (Cloud Run, Cloud Build, Artifact Registry, Firestore, Secret Manager), Git, CI/CD

## WORK EXPERIENCE

### Yourhome Platform | AI Engineer | Mar 2026 – Aug 2026

- Developing StaffLenz AI, an edge-first computer vision platform for workplace analytics, integrating multi-camera RTSP streams, YOLOv11-Pose, InsightFace/ArcFace, OpenVINO FP16 inference, polygon desk zones, and a latest-frame batching queue for real-time processing.
- Improved worker identity stability using face embeddings, head-pose filtering, identity caching, and multi-angle enrollment; connected recognized workers with desk-zone and activity-state logic.
- Implemented real-time SSE event delivery, activity history, retention cleanup, and LINE alert workflows to connect computer vision events with the management dashboard.
- Developed an asynchronous Thai speech/text-to-search pipeline using Groq STT, Llama/Gemini, Pydantic validation, and fuzzy location normalization to convert unstructured inquiries into structured search parameters.
- Built a config-driven AI-assisted QA framework using Playwright, Gemini, and Python to generate matrix test cases, validate API behavior against database rules, and surface backend regressions.
- Configured Docker-based CI/CD workflows with Google Cloud Build, Artifact Registry, and Cloud Run to prepare backend and management services for future cloud deployment.

### Sky Visual Imaging Venture Company Ltd. (SKYVIV) | AI & Geospatial Intern | May 2024 - Jul 2024

- Applied YOLO and OpenCV to drone imagery for structural crack detection in pipe inspection, and analyzed LiDAR and multispectral data for 3D and computer vision workflows.
- Developed Random Forest and Gradient Boosting regression models in MATLAB to forecast crop yield from geospatial data.

## SELECTED PROJECTS

### Multi-Country License Plate Recognition (Thai & Lao) API | Personal Project | Aug 2025 – Aug 2026

- Engineered an enterprise Multi-Country LPR system (Thai & Lao) using 100% commercially permissive models (Apache-2.0/MIT); implemented 2-stage MobileNetV3 4-corner keypoint rectification, RF-DETR character localization, and dual-engine OCR (50-class MobileNetV2 at 99.58% Top-1 and ResNet-CTC) with a compact grayscale ResNet18 classifier achieving 99.20% Top-1 across 77 Thai provinces; enforced DLT legal syntax, deployed the Dockerized FastAPI system to GCP Cloud Run (~75–95ms CPU latency), and persisted results via async workers to Firestore and BigQuery.

### LINE Stock Analysis AI Agent | Personal Project | Dec 2025 - Feb 2026

- Built a Gemini 2.5 Flash assistant integrating Thai and US market data, technical indicators, news, user risk profiles, PostgreSQL/SQLAlchemy, scheduled alerts, webhook deduplication, and asynchronous background processing.

### Hybrid RAG & Thai Legal Retrieval | Personal Projects | 2025 - 2026

- Implemented document Q&A with LangChain, ChromaDB, multilingual embeddings, and Gemini/Ollama; built a Thai legal retriever using dense search, BM25, RRF, metadata filters, query expansion, and LLM-as-a-judge evaluation.

## EDUCATION

### Chulalongkorn University, Thailand | B.Eng. Survey Engineering | 2021 – 2025

Relevant Coursework: Artificial Intelligence for Engineers; Geospatial Data Science and Analysis

## COMPETITIONS & ACTIVITIES

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- **Hackathon:** Designed a waste management solution in a sustainability-focused competition.
- **TheChallengerXKrungsri:** Developed a FinTech value proposition tailored for Gen Z users.